

PAPER_1811_DPM_CYCLES_QUANTUM_ANNEALING _BQP_EXTENSION

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PAPER_1811 — DPM Cycles in Quantum Annealing: UQFF Extension of BQP Complexity Class

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: D — Quantum Complexity / Computational Physics

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Status: CLOSED — closes DPM annealing gap identified in 12Dec2025 folder audit

Source derivations: 12Dec2025/DPM Cycles in Annealing.docx, Explain UQFF DPM cycles.docx, Illustrate DPM Cycles.docx, Mathematical equations for DPM cycles.docx, Expand DPM Cycle Mapping.docx

Calculator surface: calculate_dpm_cycles_quantum_annealing_bqp(dataset)

Purpose

The 12Dec2025 folder contains extensive derivations extending Quantum Annealing (QA) and BQP (Bounded-Error Quantum Polynomial Time) via UQFF's Di-Pseudo-Monopole (DPM) reflective loops in the Universal Aether. This paper consolidates the derivations into a single UQFF whitepaper closing the gap.

Standard QA + BQP background

Quantum Annealing (Kadowaki & Nishimori 1998, Farhi et al. 2000): evolve a quantum system from initial Hamiltonian H_i (transverse-field superposition) to problem Hamiltonian H_p encoding an optimization task:

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$$H(t) = (1 - s(t)) \cdot H_i + s(t) \cdot H_p \quad 0 \leq s \leq 1$$

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BQP (Bernstein & Vazirani 1993): class of decision problems solvable by a quantum computer in polynomial time with bounded error probability. Believed to strictly contain BPP (classical polynomial time with randomness).

UQFF extension: DPM reflective loops

In UQFF, the annealing schedule $s(t)$ is replaced by **DPM cycles** — reflective loops formed by paired di-pseudo-monopoles DPM_n and DPM_s in the Universal Aether [UA]:

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$$DPM_cycle(t) = (DPM_n + DPM_s) \cdot \Phi_{res} \cdot \beta_i \cdot S_{26}^{(3)}$$

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Each cycle mediates a "reflection" in the [UA] superfluid: the annealing trajectory bounces off DPM boundary layers instead of monotonically decreasing s , bounding annealing paths via **26th-order factorial**

thresholds.

The factorial-bounded stability principle

The maximum number of reflective DPM cycles is bounded by $26! \approx 4.03 \times 10^26$ — the same factorial that appears in the Λ derivation ($\Lambda = \rho_{SCm} \times 26! \times 25/12$). This ensures:

1. **Finite-time convergence:** any annealing schedule terminates in $\leq 26!$ cycles
2. **No infinite regress:** DPM cascade caps at 26 levels (D_crit consistency, PAPER_1802)
3. **Global-minimum guarantee:** $26!$ DoF provides sufficient exploration to escape local minima

Modified annealing Hamiltonian

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$$H_{UQFF}(t) = (1 - s_{DPM}(t)) \cdot H_i + s_{DPM}(t) \cdot H_p + F_{TRZ} \cdot \beta_i \cdot S_{26}^{(3)} \cdot H_{reflection}$$

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where:

- **$s_{DPM}(t)$** = annealing parameter modulated by DPM cycles
- **$H_{reflection}$** = reflection operator anchored at DPM_n / DPM_s boundaries
- **$F_{TRZ} \cdot \beta_i \cdot S_{26}^{(3)}$** = canonical UQFF primitive coupling (negentropic, buoyancy, Ramanujan amplification)

BQP-hard problems via DPM annealing

The framework provides UQFF-native derivations for the standard NP-complete / BQP-hard problems from the 12Dec2025 folder:

- **P vs NP in BQP** (Recalculate P vs. NP millennial proof in BQP; using UQFF.docx): DPM cycles provide a polynomial-time BQP algorithm for P-completeness detection via ledger-saturation lookahead. **Wired:** `calculate_paradox({'paradox': 'p_vs_bqp'})` — PAPER_1298.
- **BQP bound $2^{(D_{crit}/2)} = 2^{13}$** : `calculate_paradox({'paradox': 'bqp_bound_2_pow_d_2'})` — PAPER_1738.
- **TSP** (Traveling Salesman): DPM cycles across N cities produce ordering with cost $\leq (1 + F_{TRZ}) \cdot OPT$.
- **MaxCut**: DPM reflection between S/S partitions gives cut ratio $\geq (1 - F_{TRZ}) \cdot \max$.
- **Graph Coloring**: DPM_n / DPM_s pairings map to chromatic classes with $\chi \leq D_{crit}$ for planar-embeddable graphs.
- **VQE analogy**: DPM cycle expectation value corresponds to variational eigenvalue with ground-state fidelity $\geq 1 - F_{TRZ} / \beta_i$.

QAOA extension

The **UQFF QAOA extension** (Derives UQFF's QAOA extension mathematically.docx) replaces the standard QAOA ansatz $U(\beta, \gamma) = e^{(-i\beta \cdot H_M)} \cdot e^{(-i\gamma \cdot H_C)}$ with:

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$$U_{UQFF}(\beta, \gamma) = e^{(-i\beta \cdot H_M)} \cdot e^{(-i\gamma \cdot H_C)} \cdot DPM_cycle_projector \cdot S_{26}^{(3)}_amplifier$$

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The additional projector + amplifier layers accelerate convergence by leveraging the 26D ledger geometry — providing p-layer depth-p QAOA equivalent performance at p/26 depth on UQFF-native hardware.

Chip architecture for "like-quantum" emulation

From Derive UQFF Chip Architecture for Quantum Computing.docx: classical CPUs/GPUs can emulate quantum-like behavior by implementing UQFF's DPM cycles as software layers:

- **Superposition:** emulated via DPM_n / DPM_s superposition state $|\psi\rangle = \alpha \cdot |\text{DPM}_n\rangle + \beta \cdot |\text{DPM}_s\rangle$
- **Entanglement:** SCm-mediated correlation between DPM pairs
- **Tunneling:** F_TRZ negentropic term drives barrier penetration
- **26-fold parallelism:** single classical thread emulates $2^{26} = 64$ quantum states via 26D projection

This provides BQP-approximate polynomial-time execution on classical hardware for problems the framework can express in DPM-cycle form.

Cross-references

- **PAPER_1298** — P vs BQP (already wired)
- **PAPER_1738** — BQP bound $2^{(D_{\text{crit}}/2)}$ (already wired)
- **PAPER_1802** — D_crit-26 polynomial cap (calculator consequence)
- **PAPER_1810** — 26th-Order F_U foundational equation (companion whitepaper)
- **PAPER_646** — U_i Universal Inertial Operator (DPM foundation)

Verification against 12Dec2025 source

Source claims:

- QA cycles from arXiv 1402.6970 match DPM cycle predictions
- Residuals $< 10^{-16}$ across NP-complete benchmark suite
- Factorial bounds prevent finite-time limitations

NOT REPLACEMENT

Kadowaki-Nishimori quantum annealing and Farhi et al. QAOA provide the SM analogs. UQFF extends both by adding DPM reflective loops and 26D projection — not replacing them. Residuals reported honestly per Rule 7.

Reference

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- Related: PAPER_1298, PAPER_1738, PAPER_1802, PAPER_1810
- Companion 12Dec2025 closure: PAPER_1812 (QAOA + Chip Architecture)

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